

IT3091 – Machine Learning Assignment

Sri Lanka Institute of Information Technology (SLIIT)

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Project Title: Ames Housing Price Prediction

Student Name: Mummullage B.U.T

IT Number: IT24102699

1. Executive Summary

This project aims to predict the sale prices of residential properties in Ames, Iowa, using historical real estate data. The business objective was twofold:

1. Build an accurate supervised regression model to predict property prices.
2. Perform a valuation analysis to determine which property characteristics most strongly influence real estate prices.

By implementing a robust machine learning pipeline and evaluating multiple algorithms, the project successfully identified a **Tuned Gradient Boosting Regressor** as the most accurate model, achieving an R^2 score of ~ 0.91 on unseen validation data.

2. Data Exploration and Preparation

The raw dataset consisted of 1,460 observations and 79 features (both numerical and categorical). Initial Exploratory Data Analysis (EDA) revealed a right-skewed target variable (SalePrice) and highlighted strong linear relationships between price and features like GrLivArea (Above Ground Living Area) and OverallQual (Overall Quality).

Data cleaning and preprocessing steps included:

- **Outlier Removal:** Elimination of properties over 4,000 sq ft that sold for abnormally low prices to prevent linear model distortion.
- **Feature Engineering:** Creation of new composite metrics, including TotalSF (total square footage), PropertyAge (age at time of sale), and TotalBaths.
- **Imputation & Encoding:** Using scikit-learn Pipelines, numerical missing values were imputed using the median, while categorical missing values were imputed with a constant 'None'. Categorical features were subsequently one-hot encoded to prevent ordinality assumptions.

To rigorously prevent data leakage, an 80/20 train-validation split was applied prior to any data transformation.

3. Model Implementation and Comparison

The project evaluated the following algorithms:

1. **Dummy Regressor (Baseline):** Established a minimum performance threshold by predicting the mean price.

2. **Linear Regression:** Served as an interpretable benchmark model.
3. **Random Forest Regressor:** An ensemble approach utilizing bagging to reduce variance.
4. **Extra Trees Regressor:** A highly randomized tree ensemble.
5. **Gradient Boosting Regressor:** A sequential ensemble approach that builds trees to correct the errors of previous trees.

All models were evaluated using K -Fold Cross-Validation ($K=5$) on the training set and further validated against the 20% holdout set using three primary metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

4. Results and Hyperparameter Tuning

Gradient Boosting emerged as the strongest candidate. To maximize its performance, GridSearchCV was applied to tune hyperparameters including `n_estimators`, `learning_rate`, and `max_depth`.

The final tuned Gradient Boosting model achieved:

- **MAE:** ~\$14,000 (Average absolute error in predicted price)
- **R^2 :** ~0.91 (Explaining 91% of the variance in housing prices)

5. Valuation Feature Analysis

Analyzing the feature importances extracted from the tuned model provided critical business insights. The model relied heavily on:

1. **Total Square Footage (TotalSF)**
2. **Overall Material & Finish Quality (OverallQual)**
3. **Age of the Property (PropertyAge)**

This confirms standard real estate intuition: the size, condition, and age of a home are the primary drivers of its market value.

6. Limitations and Conclusion

While highly accurate, the model is subject to standard limitations. It assumes historical market conditions remain constant and cannot account for macroeconomic shifts. Furthermore, its geographic applicability is strictly limited to Ames, Iowa.

In conclusion, the project successfully demonstrated a complete end-to-end machine learning workflow, resulting in an accurate predictive model capable of assisting real estate agents in automated property benchmarking.